

Multiple Choice

1. **Answer:** A

Options: A) A group of cookies stored by the user's Web browser; B) The Internet Protocol (IP) address of the user's computer; C) The user's e-mail address; D) The user's public key used for encryption

Note: A group of cookies stored by the user's web browser can track online behavior and preferences, potentially revealing sensitive personal information to advertisers and third parties, thereby significantly compromising user privacy.

2. **Answer:** D

Options: A) 1; B) 6; C) 4; D) 5

Note: The correct answer is 5 because in Python, the modulo operator (%) has higher precedence than addition (+), so the expression is evaluated as $4 + (3 \% 2)$, which equals $4 + 1$, resulting in 5.

3. **Answer:** C

Options: A) [19,21]; B) [11,15]; C) [11,15,19]; D) [13,17,21]

Note: The correct answer is [11,15,19] because the slicing operation `b[::2]` selects every second element from the list `b`, starting from the first element at index 0.

4. **Answer:** C

Options: A) 17_{16} ; B) $E4_{16}$; C) $E7_{16}$; D) $F4_{16}$

Note: The correct answer is $E7_{16}$ because when converting the decimal number 231 to hexadecimal, it equals 14 in the 16's place (E) and 7 in the 1's place, resulting in the hexadecimal representation $E7_{16}$.

5. **Answer:** A

Options: A) $O(1)$; B) $O(n)$; C) $O(n^2)$; D) $O(\log n)$

Note: $O(1)$ is the smallest asymptotic notation because it represents a constant time complexity that does not grow with the size of the input, unlike other notations such as $O(n)$, $O(n \log n)$, or $O(n^2)$, which indicate increasing time complexities as the input size increases.

True / False

6. **Answer:** False

Options: A) True; B) False

Note: In Python 3, floor division is performed using the operator `//` instead of the operator `/`, which is used for regular division.

7. **Answer:** True

Options: A) True; B) False

Note: Authentication is compromised because it relies on verifying the identity of a user, and if an unauthorized individual gains access using stolen credentials, the system cannot accurately confirm the legitimate user's identity.

8. **Answer:** True

Options: A) True; B) False

Note: A distributed denial-of-service (DDoS) attack is characterized by the coordinated effort of multiple compromised computers, known as a botnet, that overwhelm a target's resources by sending a high volume of traffic simultaneously.

9. **Answer:** True

Options: A) True; B) False

Note: The statement is true because in Python, list slicing with the expression `list[1:3]` retrieves the elements at index 1 and index 2 from the given list, which are 786 and 2.23, resulting in the sublist `[786, 2.23]`.

10. **Answer:** True

Options: A) True; B) False

Note: In Python, variable names are considered different if they differ in capitalization, so 'Variable' and 'variable' are treated as separate identifiers due to the language's case sensitivity.

Long Form Response

11. **Answer:** `def get_total_number_of_sequences(m,n): | return T[m][n]`

Note: correct because it utilizes a dynamic programming approach by storing the computed values in a table T, where `T[m][n]` represents the total number of valid sequences of length n with the maximum value m, effectively encapsulating the problem constraints of the sequence formation.

12. **Answer:** `def concatenate_elements(test_tup): | return (res)`

Note: The answer correctly identifies the need to define a function that processes tuples, suggesting that it will return a result based on concatenating adjacent elements, which is the core requirement of the question.

End of Answer Key